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Abstract:	Let $\xi(1/2 + z) = \sum_{m=0}^{\infty} \gamma(m) z^{2m}/m!$, and define the Jensen polynomial $J_{\gamma}^{d,n}(X) = \sum_{j=0}^d \binom{d}{j} \gamma(n+j) X^j$. For every $\epsilon > 0$, we prove a subcritical real-rootedness criterion in the range $M = n + d \geq C_{\epsilon} d^{3+\epsilon}$. The proof applies the uniform Griffin-Ono-Rolen-Thorner-Tripp-Wagner normalization package and derives the exact Hermite reconstruction, together with a de-Gaussianized tube estimate, Rice finite differences, a complex shifted-saddle estimate, and a Hermite perturbation argument. The analytic Hermite reconstruction is identified exactly with the Griffin-normalized Jensen polynomial, so the final affine transfer applies to the original polynomial $J_{\gamma}^{d,n}$. The endpoint scale M asymptotic to d^3 is not treated.

Subcritical Hyperbolicity of Jensen Polynomials Associated with the Riemann Xi-Function

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May 2026

Abstract

Let

$$\xi\left(\frac{1}{2}+z\right)=\sum_{m=0}^{\infty}\gamma(m)\frac{z^{2m}}{m!},$$

and define the Jensen polynomial

$$J_{\gamma}^{d,n}(X)=\sum_{j=0}^d\binom{d}{j}\gamma(n+j)X^j.$$

For every $\varepsilon > 0$, we prove a subcritical real-rootedness criterion in the range

$$M=n+d\geq C_{\varepsilon}d^{3+\varepsilon}.$$

The proof applies the uniform Griffin–Ono–Rolen–Thorner–Tripp–Wagner normalization package and derives the exact Hermite reconstruction, together with a de-Gaussianized tube estimate, Rice finite differences, a complex shifted-saddle estimate, and a Hermite perturbation argument. The analytic Hermite reconstruction is identified exactly with the Griffin-normalized Jensen polynomial, so the final affine transfer applies to the original polynomial $J_{\gamma}^{d,n}$. The endpoint scale $M \asymp d^3$ is not treated.

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1 Introduction

The Riemann xi-function is

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

We write its Taylor expansion at the central point in the form

$$\xi\left(\frac{1}{2} + z\right) = \sum_{m=0}^{\infty} \gamma(m) \frac{z^{2m}}{m!}.$$

For integers $d, n \geq 1$, define the Jensen polynomial

$$J_{\gamma}^{d,n}(X) = \sum_{j=0}^d \binom{d}{j} \gamma(n+j) X^j.$$

Throughout this paper, a real polynomial is called *hyperbolic* if all of its zeros are real.

The Pólya–Jensen criterion [1, 2] relates the Riemann hypothesis to the hyperbolicity of Jensen polynomials attached to the Taylor coefficients of ξ . Griffin–Ono–Rolen–Zagier proved, for each fixed degree, eventual hyperbolicity of the corresponding Jensen polynomials and developed the Hermite-polynomial limiting framework [3]. Griffin–Ono–Rolen–Thorner–Tripp–Wagner later made this eventual hyperbolicity effective for the Riemann xi-function, with an effective bound that is exponential in the degree [4]. O’Sullivan developed related Hermite-expansion criteria and asymptotic formulae for the xi-coefficients [5].

The purpose of the present paper is to prove a polynomial-scale subcritical effective bound. The main theorem is the following.

Theorem 1.1 (Subcritical hyperbolicity). *For every $\varepsilon > 0$, there exists $C_{\varepsilon} > 0$ such that, whenever*

$$M = n + d \geq C_{\varepsilon} d^{3+\varepsilon},$$

the polynomial $J_{\gamma}^{d,n}(X)$ is hyperbolic.

The proof starts from the effective normalization package of Griffin–Ono–Rolen–Thorner–Tripp–Wagner. The first point is an exact reconstruction identity: the analytic Hermite reconstruction used below is not a surrogate polynomial but is exactly the Griffin-normalized Jensen polynomial. This identity fixes the Gaussian scale as $\Delta(M)^2$, not as $G_2(M)\Delta(M)^2$. After this normalization, the proof combines a de-Gaussianized tube estimate, Rice finite differences, a Newton–Poisson decomposition, a complex shifted-saddle estimate for the resulting kernels, and a Hermite perturbation argument.

The exponent $3 + \varepsilon$ comes from a subcritical disk radius $|t| \leq \rho_t M^{\theta}$ with $\theta < 1/6$. The endpoint $M \asymp d^3$, corresponding to $\theta = 1/6$, is a natural limiting scale for this method but is not treated here.

2 Parameter hierarchy

Set

$$M = n + d.$$

Fix once and for all a positive truncation constant $A > 0$; for example, one may take $A = 1$. Set

$$L = \lfloor AM^{2/3} \rfloor.$$

Fix $\varepsilon > 0$, and put

$$\theta = \frac{1}{2(3 + \varepsilon)}, \quad \alpha = \frac{1}{2} + \theta.$$

Then

$$0 < \theta < \frac{1}{6}, \quad \alpha < \frac{2}{3}.$$

Remark 2.1 (Order of parameter choices). *The constants are chosen in the following order. First fix $\varepsilon > 0$, then set*

$$\theta = \frac{1}{2(3 + \varepsilon)}, \quad \alpha = \frac{1}{2} + \theta.$$

Choose δ so that

$$\alpha - \frac{1}{2} < \delta < \frac{1 - \alpha}{2}.$$

This interval is nonempty exactly because $\alpha < 2/3$. The truncation parameter A is fixed once and for all. Next fix a disk parameter $\rho_t > 0$. Then choose the tube parameter B sufficiently large, relative to the fixed A and ρ_t , for the Rice decay in the bounded Hermite–Gaussian summation. Finally take M sufficiently large. All resulting lower bounds for M are absorbed into C_ε . No uniformity as $\varepsilon \rightarrow 0$ is asserted.

Remark 2.2 (Truncation range and Griffin integer range). *In the range $M \geq C_\varepsilon d^{3+\varepsilon}$, increasing C_ε if necessary gives both*

$$L \geq d \quad \text{and} \quad L < M - 1.$$

The first inequality ensures that the truncated Poisson transform has the correct Taylor coefficients through degree d . The second ensures that every integer value $S(j; M)$ used in the finite-difference reconstruction lies in the integer range in which the Griffin identities are applied.

Lemma 2.3 (Radius-to-scale conversion). *For every $c_\theta > 0$, there exists $C_\varepsilon > 0$ such that*

$$M \geq C_\varepsilon d^{3+\varepsilon} \implies \sqrt{d} \leq c_\theta M^\theta.$$

Proof. If $M \geq C_\varepsilon d^{3+\varepsilon}$, then

$$M^\theta \geq C_\varepsilon^\theta d^{(3+\varepsilon)\theta} = C_\varepsilon^\theta d^{1/2}.$$

Choosing $C_\varepsilon^\theta \geq c_\theta^{-1}$ proves the claim. □

3 Griffin normalization and exact reconstruction

This section records the consequences of Griffin–Ono–Rolen–Thorner–Tripp–Wagner used in the proof. We use the notation and numbering of the accepted *Advances in Mathematics* version of their paper. Their Theorem 2.2 defines and estimates the uniformizer $\Delta(M)$ and the functions $G_r(M)$, while equations (2.3)–(2.7), (4.1)–(4.2) define the normalized Jensen polynomial and its finite-difference coefficients. We use these inputs only after the specialization $M = n + d$. The items below spell out the exact consequences needed later, rather than treating the Griffin normalization as an informal black box.

Proposition 3.1 (Derived Griffin normalization package). *For all sufficiently large M , the following consequences of Griffin–Ono–Rolen–Thorner–Tripp–Wagner hold.*

(G1) *By the definition $\Delta(M)$ in [4, eq. (2.3)] and its estimate in [4, Theorem 2.2(1)],*

$$\Delta(M) > 0, \quad \Delta(M) \asymp M^{-1/2}.$$

(G2) *By [4, Theorem 2.2(2), eq. (2.4)], there are functions $G_r(M)$ such that, for integers $1 \leq j < M$,*

$$\log \frac{\gamma(M-j)}{\gamma(M)} = - \sum_{r=1}^{\infty} G_r(M) \Delta(M)^{2r-2} j^r.$$

The same theorem gives the joint uniform exponential bound

$$|G_r(M)| \leq C_0 C_1^r \quad (r \geq 1),$$

with constants independent of the joint parameters r and M . This uniformity in r is the input used to form the complex tube extension below.

(G3) *By [4, Theorem 2.2(3), eq. (2.5)],*

$$G_2(M) = 1 + O(\Delta(M)^2) = 1 + O(M^{-1}).$$

(G4) *Define, following [4, eq. (4.1)],*

$$S(j; M) = \frac{R_M(j)}{R_M(1)^j}.$$

Combining the preceding logarithmic expansion with this definition gives, for integers $0 \leq j < M$,

$$S(j; M) = \exp \left(\sum_{r=2}^{\infty} G_r(M) \Delta(M)^{2r-2} (j - j^r) \right).$$

The same exponential series defines the holomorphic extension $S(z; M)$ in the $M^{2/3}$ -tube by theorem 3.2. The holomorphic function $S(z; M)$ used below is this extension of the Griffin integer formula; the Griffin identities themselves are invoked only at integer arguments.

(G5) *By [4, eqs. (2.6)–(2.7), eq. (4.2)], after the specialization $M = n + d$, the Griffin-normalized Jensen polynomial*

$$\tilde{J}_\gamma^{d,n}(X) = \sum_{m=0}^d A_{d,m}(n) X^{d-m}$$

has coefficients

$$A_{d,m}(n) = \binom{d}{m} \Delta(M)^{-m} \mathfrak{D}^m S(0; M), \quad 0 \leq m \leq d,$$

where

$$\mathfrak{D}^m S(0; M) = \sum_{j=0}^m (-1)^{m-j} \binom{m}{j} S(j; M).$$

Moreover, the affine normalization in [4, eq. (2.6)], again with $M = n + d$, is

$$\tilde{J}_\gamma^{d,n}(X) = \frac{\gamma(M)^{d-1}}{\gamma(M-1)^d} \Delta(M)^d J_\gamma^{d,n} \left(\frac{\gamma(M-1)}{\gamma(M)} (\Delta(M)X - 1) \right).$$

Lemma 3.2 (Complex extension on the $M^{2/3}$ -tube). *Let $A, B > 0$ be fixed, and set*

$$L = \lfloor AM^{2/3} \rfloor, \quad R = BM^{2/3},$$

$$\Omega_R = \{z \in \mathbb{C} : \text{dist}(z, [0, L]) < R\}.$$

The exponential series defining $S(z; M)$ in theorem 3.1 converges absolutely and locally uniformly on Ω_R .

Proof. For $z \in \Omega_R$, $|z| \leq C_{A,B}M^{2/3}$. By theorem 3.1,

$$|G_r(M)|\Delta(M)^{2r-2}|z|^r \leq C^r M^{-(r-1)}M^{2r/3} = C^r M^{1-r/3}.$$

The term $r = 3$ is $O(1)$, and the terms $r \geq 4$ form a uniformly convergent tail for large M . The linear terms $G_r(M)\Delta(M)^{2r-2}z$ are smaller. This proves absolute and locally uniform convergence. $\square$

3.1 De-Gaussianization

The main Gaussian parameter is

$$\lambda_M := \Delta(M)^2.$$

This choice is forced by the finite-difference reconstruction in theorem 3.5. We write

$$S(z; M) = \exp(\lambda_M z - \lambda_M z^2)F_M(z), \quad F_M(z) = \exp W_M(z),$$

where

$$W_M(z) = (G_2(M) - 1)\Delta(M)^2(z - z^2) + \sum_{r=3}^{\infty} G_r(M)\Delta(M)^{2r-2}(z - z^r).$$

Lemma 3.3 (De-Gaussianized tube bound). *For fixed $A, B > 0$,*

$$\sup_{z \in \Omega_R} |F_M(z)| \leq C_{A,B}$$

for all sufficiently large M .

Proof. For $z \in \Omega_R$, $|z| \leq C_{A,B}M^{2/3}$. Since $G_2(M) - 1 = O(M^{-1})$ and $\Delta(M)^2 \asymp M^{-1}$,

$$|G_2(M) - 1|\Delta(M)^2|z|^2 \ll M^{-1}M^{-1}M^{4/3} = M^{-2/3}.$$

For $r \geq 3$,

$$|G_r(M)|\Delta(M)^{2r-2}|z|^r \leq C^r M^{1-r/3},$$

so the $r = 3$ contribution is bounded and the $r \geq 4$ tail converges uniformly. The linear parts are smaller. Hence $|W_M(z)| \leq C_{A,B}$, and therefore $|F_M(z)| \leq e^{C_{A,B}}$. $\square$

3.2 Poisson transform and exact reconstruction

Define

$$\mathcal{A}_M^{(L)}(x) = e^{-x} \sum_{j=0}^L S(j; M) \frac{x^j}{j!}, \quad A_M^{(L)}(t) = \mathcal{A}_M^{(L)}\left(\frac{t}{\Delta(M)}\right),$$

and

$$E_M^{(L)}(t) = e^{t^2} A_M^{(L)}(t).$$

Lemma 3.4 (Low-degree coefficients of the truncated Poisson transform). *If $L \geq d$, then for $0 \leq m \leq d$,*

$$[x^m] \mathcal{A}_M^{(L)}(x) = \frac{1}{m!} \mathfrak{D}^m S(0; M).$$

Proof. For $m \leq L$,

$$[x^m] \left(e^{-x} \sum_{j=0}^L S(j; M) \frac{x^j}{j!} \right) = \sum_{j=0}^m S(j; M) \frac{(-1)^{m-j}}{(m-j)!j!} = \frac{\mathfrak{D}^m S(0; M)}{m!}.$$

□

Proposition 3.5 (Exact Griffin–Hermite reconstruction identity). *Assume $L \geq d$. Then*

$$d![t^d] \left(e^{Xt-t^2} E_M^{(L)}(t) \right) = \sum_{m=0}^d \binom{d}{m} \Delta(M)^{-m} \mathfrak{D}^m S(0; M) X^{d-m}.$$

Consequently this polynomial is exactly the Griffin-normalized Jensen polynomial $\tilde{J}_\gamma^{d,n}(X)$.

Proof. Since $E_M^{(L)}(t) = e^{t^2} A_M^{(L)}(t)$,

$$e^{Xt-t^2} E_M^{(L)}(t) = e^{Xt} A_M^{(L)}(t).$$

By theorem 3.4, for $m \leq d$,

$$[t^m] A_M^{(L)}(t) = \Delta(M)^{-m} \frac{\mathfrak{D}^m S(0; M)}{m!}.$$

Therefore

$$\begin{aligned} d![t^d] e^{Xt} A_M^{(L)}(t) &= d! \sum_{m=0}^d \frac{X^{d-m}}{(d-m)!} \Delta(M)^{-m} \frac{\mathfrak{D}^m S(0; M)}{m!} \\ &= \sum_{m=0}^d \binom{d}{m} \Delta(M)^{-m} \mathfrak{D}^m S(0; M) X^{d-m}. \end{aligned}$$

By theorem 3.1, this is exactly the Griffin-normalized Jensen polynomial. □

Corollary 3.6 (The truncation does not affect degree d). *In the range $M \geq C_\varepsilon d^{3+\varepsilon}$, after increasing C_ε if necessary, one has*

$$L \geq d, \quad L < M - 1.$$

Hence the truncated transform $\mathcal{A}_M^{(L)}$ gives the exact degree- d reconstruction, and all integer values $S(j; M)$ used in the finite differences lie in the Griffin integer range.

Proof. The hypothesis gives $d \ll M^{1/(3+\varepsilon)}$, while $L \asymp M^{2/3}$. Since $1/(3+\varepsilon) < 2/3$, increasing C_ε ensures $L \geq d$. Also $L = O(M^{2/3}) < M - 1$ for all sufficiently large M . The finitely many excluded values are absorbed by increasing C_ε . □

4 Rice bound and Newton–Poisson decomposition

Lemma 4.1 (Rice finite-difference bound). *Assume F_M is holomorphic in Ω_R and satisfies*

$$\sup_{z \in \Omega_R} |F_M(z)| \leq C_F.$$

Then for $0 \leq k \leq L$,

$$\left| \frac{\mathfrak{D}^k F_M(0)}{k!} \right| \leq C R^{-k}.$$

Proof. Rice's formula gives

$$\mathfrak{D}^k F_M(0) = \frac{k!}{2\pi i} \int_{\Gamma_k} \frac{F_M(z)}{z(z-1)\cdots(z-k)} dz,$$

where Γ_k surrounds the integers $0, 1, \dots, k$. We choose Γ_k to be the positively oriented boundary of the rectangle

$$-\frac{R}{4} \leq \Re z \leq k + \frac{R}{4}, \quad |\Im z| \leq \frac{R}{4}.$$

This rectangle is contained in Ω_R for every $0 \leq k \leq L$. Indeed, if $0 \leq \Re z \leq k \leq L$, then $\text{dist}(z, [0, L]) \leq R/4$. If $\Re z < 0$, then $-R/4 \leq \Re z < 0$ and $|\Im z| \leq R/4$, so $\text{dist}(z, [0, L]) \leq R/\sqrt{8} < R$. The same bound holds on the right side because $k \leq L$ and $k < \Re z \leq k + R/4$. Hence the contour lies strictly inside the tube, with a fixed numerical margin.

For every $z \in \Gamma_k$ and every $0 \leq j \leq k$, at least one of the horizontal or vertical separations from j is $\gg R$; hence

$$|z - j| \geq cR, \quad 0 \leq j \leq k,$$

with $c > 0$ depending only on the fixed shape of the rectangle. Thus

$$|z(z-1)\cdots(z-k)| \geq (cR)^{k+1}.$$

The length of the rectangle is

$$O(k + R) = O_{A,B}(R),$$

since $k \leq L \asymp_A M^{2/3}$ and $R = BM^{2/3}$. Therefore

$$|\mathfrak{D}^k F_M(0)| \leq Ck! C_F R (cR)^{-(k+1)} \leq Ck! R^{-k},$$

which proves the claim. □

Lemma 4.2 (Newton–Poisson decomposition). *Let*

$$\mathfrak{D}^k F_M(0) = \sum_{r=0}^k (-1)^{k-r} \binom{k}{r} F_M(r).$$

Then

$$\mathcal{A}_M^{(L)}(x) = \sum_{k=0}^L \frac{\mathfrak{D}^k F_M(0)}{k!} x^k \mathcal{K}_{k,L}(x),$$

where

$$\mathcal{K}_{k,L}(x) = e^{-x} \sum_{\ell=0}^{L-k} \frac{x^\ell}{\ell!} \exp(\lambda_M(k+\ell) - \lambda_M(k+\ell)^2),$$

and $\lambda_M = \Delta(M)^2$.

Proof. Substitute the finite Newton expansion

$$F_M(j) = \sum_{k=0}^j \binom{j}{k} \mathfrak{D}^k F_M(0)$$

into $\mathcal{A}_M^{(L)}$, interchange finite sums, and set $j = k + \ell$. The identity $j!^{-1} \binom{j}{k} = (k!\ell!)^{-1}$ gives the formula. □

5 The shifted-saddle kernel

For this section write $\lambda = \Delta(M)^2$. The variable x is complex throughout this section. All estimates below are upper bounds in terms of $|x|$ and real parts of phases; no positivity or reality of x is used. Let $0 \leq k \leq L$, $N = L - k$, and define

$$\mathcal{K}_{k,L}(x) = e^{-x} \sum_{\ell=0}^N \frac{x^\ell}{\ell!} \exp(\lambda(k+\ell) - \lambda(k+\ell)^2).$$

Proposition 5.1 (Complex shifted-saddle kernel estimate). *Fix $\rho > 0$. There exist constants $C_\rho, c_\rho, C_* > 0$, depending on the fixed parameters A, ρ, α but not on M, k, L, N, x , such that the following holds. Assume*

$$|x| \leq \rho M^\alpha, \quad \alpha < \frac{2}{3}.$$

Put $t = \sqrt{\lambda}x$. In the main range

$$N \geq C_*(1 + |x|),$$

one has

$$|e^{t^2} \mathcal{K}_{k,L}(x)| \leq C_\rho \exp\left(C_\rho \rho \frac{k}{M^{1-\alpha}} + C_\rho \frac{k^2}{M}\right) + C_\rho e^{-c_\rho N}.$$

In the boundary range $N < C_*(1 + |x|)$,

$$|e^{t^2} \mathcal{K}_{k,L}(x)| \leq \exp(C_\rho M^\alpha).$$

5.1 Infinite kernel and shifted contour

Define

$$\mathcal{K}_{k,\infty}(x) = e^{-x} \sum_{\ell=0}^{\infty} \frac{x^\ell}{\ell!} \exp(\lambda(k+\ell) - \lambda(k+\ell)^2).$$

Lemma 5.2 (Fourier–Gaussian representation). *For $\lambda > 0$,*

$$\mathcal{K}_{k,\infty}(x) = e^{\lambda k} \frac{1}{\sqrt{4\pi\lambda}} \int_{\mathbb{R}} \exp\left(-\frac{u^2}{4\lambda} + iuk + x(e^{\lambda+iu} - 1)\right) du.$$

Proof. Insert

$$e^{-\lambda n^2} = \frac{1}{\sqrt{4\pi\lambda}} \int_{\mathbb{R}} e^{-u^2/(4\lambda) + iun} du$$

with $n = k + \ell$. Since $x \in \mathbb{C}$, we justify the interchange of the ℓ -sum and the u -integral by absolute convergence:

$$\sum_{\ell=0}^{\infty} \frac{(|x|e^\lambda)^\ell}{\ell!} \int_{\mathbb{R}} e^{-u^2/(4\lambda)} du = e^{e^\lambda |x|} \sqrt{4\pi\lambda} < \infty.$$

Thus Fubini's theorem gives

$$\begin{aligned} \mathcal{K}_{k,\infty}(x) &= e^{-x+\lambda k} \frac{1}{\sqrt{4\pi\lambda}} \int_{\mathbb{R}} e^{-u^2/(4\lambda) + iuk} \sum_{\ell=0}^{\infty} \frac{(xe^{\lambda+iu})^\ell}{\ell!} du \\ &= e^{\lambda k} \frac{1}{\sqrt{4\pi\lambda}} \int_{\mathbb{R}} \exp\left(-\frac{u^2}{4\lambda} + iuk + x(e^{\lambda+iu} - 1)\right) du. \end{aligned}$$

□

After multiplying by $e^{\lambda x^2}$, define

$$\Phi(u) = -\frac{u^2}{4\lambda} + iuk + x(e^{\lambda+iu} - 1) + \lambda x^2.$$

Lemma 5.3 (Saddle point and expanded saddle value). *The equation $\Phi'(u) = 0$ has a unique solution u_s satisfying*

$$|u_s| \leq C\lambda(k + |x|).$$

Moreover,

$$\Re\Phi(u_s) \leq C + C\lambda k|x| + C\lambda k^2,$$

where the constant is uniform for $0 \leq k \leq L$ and $|x| \leq \rho M^\alpha$, with fixed ρ .

Proof. The saddle equation is

$$u = 2i\lambda(k + xe^{\lambda+iu}).$$

On the disk $|u| \leq C\lambda(k + |x|)$, the right side maps the disk into itself for C large. Its derivative is bounded by

$$2\lambda|x|e^{\lambda+|\Im u|} = o(1),$$

since $\lambda|x| \ll M^{\alpha-1} = o(1)$ and $|u| = O(M^{-1/3})$. Thus the contraction principle gives a unique saddle and

$$|u_s| \leq C\lambda(k + |x|).$$

Put

$$u_0 = 2i\lambda(k + x), \quad u_s = u_0 + E_s.$$

From the saddle equation and $e^{\lambda+iu_s} = 1 + O(\lambda + |u_s|)$,

$$E_s = O(\lambda|x|(\lambda + |u_s|)) = O(\lambda^2|x|(1 + k + |x|)).$$

We expand the phase term by term. First,

$$-\frac{u_s^2}{4\lambda} + iu_s k = -\frac{u_0^2}{4\lambda} + iu_0 k + O\left(\frac{|u_0 E_s|}{\lambda} + \frac{|E_s|^2}{\lambda} + k|E_s|\right).$$

Since $u_0 = 2i\lambda(k + x)$,

$$-\frac{u_0^2}{4\lambda} + iu_0 k = \lambda(k + x)^2 - 2\lambda k(k + x) = \lambda x^2 - \lambda k^2.$$

Next,

$$x(e^{\lambda+iu_s} - 1) = x(\lambda + iu_s) + O(|x|(\lambda + |u_s|)^2),$$

and

$$x(\lambda + iu_s) = \lambda x - 2\lambda kx - 2\lambda x^2 + O(|x||E_s|).$$

Adding the external term λx^2 , the main part is

$$\lambda x - \lambda k^2 - 2\lambda kx.$$

Therefore

$$\Phi(u_s) = \lambda x - \lambda k^2 - 2\lambda kx + \mathcal{E},$$

with

$$|\mathcal{E}| \leq C\left(\frac{|u_0 E_s|}{\lambda} + \frac{|E_s|^2}{\lambda} + k|E_s| + |x||E_s| + |x|(\lambda + |u_s|)^2\right).$$

We now record the sizes of the error terms. Since $|u_0| \leq C\lambda(k + |x|)$,

$$\frac{|u_0 E_s|}{\lambda} \ll \lambda^2 |x| (k + |x|) (1 + k + |x|),$$

and hence it is bounded by a finite sum of the model terms

$$|x|\lambda^2 k^2, \quad |x|\lambda^2 k|x|, \quad |x|\lambda^2 |x|^2, \quad |x|\lambda^2.$$

Likewise

$$k|E_s| + |x||E_s| \ll \lambda^2 |x| (k + |x|) (1 + k + |x|).$$

For the square term, the bound for E_s gives

$$\frac{|E_s|^2}{\lambda} \ll \lambda^3 |x|^2 (1 + k + |x|)^2.$$

Since $\lambda|x| = o(1)$, this is dominated by

$$\lambda^2 |x| (1 + k + |x|)^2,$$

and hence by the same model terms above. Finally,

$$|x|(\lambda + |u_s|)^2 \ll |x|\lambda^2 (1 + k + |x|)^2,$$

which is controlled by the same model terms. These model terms satisfy

$$|x|\lambda^2 k^2 \ll M^{\alpha-2/3} = O(1), \quad |x|\lambda^2 k|x| \ll M^{2\alpha-4/3} = O(1),$$

$$|x|\lambda^2 |x|^2 \ll M^{3\alpha-2} = O(1), \quad |x|\lambda^2 \ll M^{\alpha-2} = o(1),$$

because $k \leq AM^{2/3}$, $|x| \leq \rho M^\alpha$, $\lambda \asymp M^{-1}$, and $\alpha < 2/3$. Thus $|\mathcal{E}| = O(1)$, uniformly in the stated range. Taking real parts,

$$\Re \Phi(u_s) \leq C + C\lambda k^2 + C\lambda k|x| + \lambda|x|,$$

and $\lambda|x| = o(1)$. This proves the displayed estimate. $\square$

Lemma 5.4 (Global shifted contour). *One has*

$$\int_{\mathbb{R}} e^{\Phi(u)} du = \int_{u_s + \mathbb{R}} e^{\Phi(u)} du.$$

Proof. The integrand is entire. Apply Cauchy's theorem to the contour formed by the segments $[-T, T]$, $[T, T + u_s]$, $[T + u_s, -T + u_s]$, and $[-T + u_s, -T]$. On the two connecting sides $T + su_s$ and $-T + su_s$, $0 \leq s \leq 1$, the real part of the Gaussian term satisfies, for T large,

$$\Re \left(-\frac{(\pm T + su_s)^2}{4\lambda} \right) \leq -\frac{T^2}{8\lambda} + C \frac{|u_s|^2}{\lambda}.$$

Moreover, on the connecting sides one has

$$|e^{\lambda + i(\pm T + su_s)}| = e^{\lambda - \Im(su_s)} \leq e^{\lambda + |u_s|} = O(1),$$

so

$$|x(e^{\lambda + i(\pm T + su_s)} - 1)| \leq C|x|,$$

with constants independent of T . Thus the connecting-side integrals are bounded by

$$O \left(|u_s| \exp \left(-\frac{T^2}{8\lambda} + C \frac{|u_s|^2}{\lambda} + C|x| + \lambda|x|^2 \right) \right),$$

which tends to zero as $T \rightarrow \infty$. The horizontal integrals therefore converge to the two integrals in the statement. $\square$

Lemma 5.5 (Global shifted-line decay). *For $v \in \mathbb{R}$,*

$$\Re \Phi(u_s + v) \leq -c \frac{v^2}{\lambda} + C\lambda(k + |x|)^2 + C|x| + C\lambda|x|^2.$$

Consequently, for each fixed $u_0 > 0$, the far range $|v| > u_0$ contributes $O(e^{-c'M})$ after the Fourier prefactor is included.

Proof. For the Gaussian part,

$$\begin{aligned} \Re \left(-\frac{(u_s + v)^2}{4\lambda} \right) &\leq -c \frac{v^2}{\lambda} + C \frac{|u_s|^2}{\lambda} \\ &\leq -c \frac{v^2}{\lambda} + C\lambda(k + |x|)^2, \end{aligned}$$

because $|u_s| \leq C\lambda(k + |x|)$. Since $v \in \mathbb{R}$,

$$|e^{\lambda + i(u_s + v)}| \leq e^{\lambda + |\Im u_s|} = O(1),$$

and hence

$$|x(e^{\lambda + i(u_s + v)} - 1)| \leq C|x|$$

uniformly in v . Adding the external term λx^2 gives the displayed bound. If $|v| > u_0$, then $v^2/\lambda \geq cM$, while

$$\lambda(k + |x|)^2 + |x| + \lambda|x|^2 = o(M)$$

under $k \leq AM^{2/3}$, $|x| \leq \rho M^\alpha$, and $\alpha < 2/3$. Hence the far range is $O(e^{-cM + o(M)}) = O(e^{-c'M})$. The factor $\lambda^{-1/2}$ is absorbed by this exponential decay. $\square$

Lemma 5.6 (Quadratic decrease near the saddle). *There exists a fixed $u_0 > 0$ such that for $|v| \leq u_0$,*

$$\Re(\Phi(u_s + v) - \Phi(u_s)) \leq -c \frac{v^2}{\lambda}.$$

Proof. Taylor's formula gives

$$\Phi(u_s + v) - \Phi(u_s) = \frac{1}{2} \Phi''(u_s) v^2 + O(|x||v|^3),$$

where

$$\Phi''(u) = -\frac{1}{2\lambda} - x e^{\lambda + iu}, \quad \Phi'''(u) = -i x e^{\lambda + iu}.$$

Indeed, for $|w - u_s| \leq u_0$, with u_0 fixed, one has

$$|\Phi'''(w)| \leq |x| e^{\lambda + |\Im w|} \leq C|x|,$$

because $|u_s| = o(1)$. Since $\lambda|x| = o(1)$, the real part of the quadratic coefficient is $\leq -c/\lambda$. Fix $u_0 > 0$ sufficiently small. Since $\lambda|x| = o(1)$ uniformly in the present range, for all sufficiently large M the cubic term is dominated by the quadratic term on $|v| \leq u_0$. Indeed, for $|v| \leq u_0$,

$$|x||v|^3 \leq (\lambda|x|u_0) \frac{v^2}{\lambda} = o\left(\frac{v^2}{\lambda}\right).$$

Thus the negative quadratic term dominates. $\square$

Lemma 5.7 (Infinite shifted-kernel estimate with prefactor). *Under the assumptions of theorem 5.1,*

$$|e^{\lambda x^2} \mathcal{K}_{k,\infty}(x)| \leq C \exp(C\lambda k|x| + C\lambda k^2).$$

Proof. By theorems 5.2 and 5.4,

$$e^{\lambda x^2} \mathcal{K}_{k,\infty}(x) = e^{\lambda k} \frac{1}{\sqrt{4\pi\lambda}} \int_{\mathbb{R}} e^{\Phi(u_s+v)} dv.$$

Choose

$$U = \lambda^{1/2} M^\delta, \quad \alpha - \frac{1}{2} < \delta < \frac{1-\alpha}{2}.$$

Since $U = M^{-1/2+\delta+o(1)} \rightarrow 0$, we may assume $U < u_0$ for all sufficiently large M .

For $|v| \leq U$, theorems 5.3 and 5.6 gives

$$\begin{aligned} \frac{1}{\sqrt{4\pi\lambda}} \int_{|v| \leq U} |e^{\Phi(u_s+v)}| dv &\leq C e^{\Re \Phi(u_s)} \frac{1}{\sqrt{\lambda}} \int_{\mathbb{R}} e^{-cv^2/\lambda} dv \\ &\leq C \exp(C\lambda k|x| + C\lambda k^2). \end{aligned}$$

Here the prefactor $\lambda^{-1/2}$ is exactly absorbed by the Gaussian width $\lambda^{1/2}$.

For $U < |v| \leq u_0$, the same quadratic decrease yields

$$C\lambda^{-1/2} \exp(C\lambda k|x| + C\lambda k^2 - cU^2/\lambda).$$

Since $U^2/\lambda = M^{2\delta}$ and $\lambda^{-1/2} \leq e^{o(M^{2\delta})}$, this is

$$C \exp(C\lambda k|x| + C\lambda k^2 - cM^{2\delta}).$$

For $|v| > u_0$, theorem 5.5 gives $O(e^{-c'M})$. Combining the three ranges and using $e^{\lambda k} = O(1)$ proves the lemma. $\square$

5.2 Truncation and boundary range

Define

$$\mathcal{R}_{k,N}(x) = \mathcal{K}_{k,\infty}(x) - \mathcal{K}_{k,L}(x).$$

Lemma 5.8 (Truncated exponential tail). *There are constants $C_*, c > 0$ such that, if $N \geq C_*(1+X)$, then*

$$\sum_{\ell > N} \frac{X^\ell}{\ell!} \leq e^{-cN}.$$

Proof. Choose C_* so large that $eX/(N+1) \leq q < 1$. Stirling's lower bound gives the first term $\leq q^{N+1}$, and the remaining terms form a geometric tail. $\square$

Lemma 5.9 (Truncation tail). *If $N \geq C_*(1+|x|)$, then*

$$|e^{\lambda x^2} \mathcal{R}_{k,N}(x)| \leq C e^{-cN}.$$

Proof. Taking absolute values and using $\lambda y - \lambda y^2 \leq C$ for $y \geq 0$,

$$|e^{\lambda x^2} \mathcal{R}_{k,N}(x)| \leq C e^{\lambda|x|^2+|x|} \sum_{\ell > N} \frac{|x|^\ell}{\ell!}.$$

By theorem 5.8, the sum is $\leq e^{-c_1 N}$. Since $N \geq C_*(1+|x|)$, we have $|x| \leq N/C_*$. Also $N \leq L \asymp M^{2/3}$, so

$$\lambda|x|^2 \leq C\lambda N^2 \leq CNM^{-1/3} = o(N).$$

Taking C_* large and then M large, the term $\lambda|x|^2 + |x|$ is absorbed into $-c_1 N$. $\square$

Lemma 5.10 (Boundary crude estimate). *If $N < C_*(1 + |x|)$, then*

$$|e^{\lambda x^2} \mathcal{K}_{k,L}(x)| \leq \exp(CM^\alpha).$$

Proof. Taking absolute values,

$$|e^{\lambda x^2} \mathcal{K}_{k,L}(x)| \leq C e^{\lambda|x|^2 + 2|x|} \leq \exp(CM^\alpha),$$

because $|x| \leq \rho M^\alpha$ and $\lambda|x|^2 \ll M^\alpha$. □

Proof of theorem 5.1. Write $\mathcal{K}_{k,L} = \mathcal{K}_{k,\infty} - \mathcal{R}_{k,N}$. The main range follows from theorems 5.7 and 5.9. Since $\lambda \asymp M^{-1}$ and $|x| \leq \rho M^\alpha$,

$$\lambda k|x| \ll \rho \frac{k}{M^{1-\alpha}}, \quad \lambda k^2 \ll \frac{k^2}{M}.$$

The boundary range follows from theorem 5.10. □

6 The complex BHG bound

Lemma 6.1 (Summing the truncation-remainder contribution). *Let*

$$a := \frac{|x|}{R} \leq \frac{C_{\rho_t}}{B} M^{-\sigma}, \quad \sigma = \frac{2}{3} - \alpha > 0.$$

For B sufficiently large and then M sufficiently large,

$$\sum_{k=0}^L a^k e^{-c(L-k)} \leq C.$$

In fact this sum tends to zero as $M \rightarrow \infty$.

Proof. Split the sum into $k \leq L/2$ and $k > L/2$. If $k \leq L/2$, then $e^{-c(L-k)} \leq e^{-cL/2}$, so this part is bounded by

$$e^{-cL/2} \sum_{k=0}^{\infty} a^k = O(e^{-cL/2}),$$

since $a < 1/2$ for large M . If $k > L/2$, then

$$a^k \leq a^{L/2} \leq \exp(-c_1 L \log M)$$

after B is fixed large and M is large. This part is at most

$$L \exp(-c_1 L \log M) = o(1).$$

□

Proposition 6.2 (Subcritical BHG bound). *For fixed $\rho_t > 0$,*

$$\sup_{|t| \leq \rho_t M^\theta} |e^{t^2} A_M^{(L)}(t)| \leq C_{\theta, \rho_t}.$$

The dependence on the fixed disk parameter ρ_t is retained in the notation and is used explicitly in the Hermite perturbation step.

Proof. Let $x = t/\Delta(M)$. Then $|x| \leq C\rho_t M^\alpha$. By theorems 4.1 and 4.2,

$$|e^{t^2} A_M^{(L)}(t)| \leq C \sum_{k=0}^L \left(\frac{|x|}{R} \right)^k |e^{t^2} \mathcal{K}_{k,L}(x)|.$$

Since $R = BM^{2/3}$,

$$\frac{|x|}{R} \leq \frac{C\rho_t}{B} M^{\alpha-2/3} = \frac{C\rho_t}{B} M^{-\sigma}, \quad \sigma = \frac{2}{3} - \alpha > 0.$$

Choose B large after fixing ρ_t , then choose M large.

In the main range, theorem 5.1 gives a principal contribution with factor

$$\left(\frac{|x|}{R} \right)^k \exp(C\lambda k|x| + C\lambda k^2).$$

This is bounded by

$$\exp(-ck \log M),$$

because $\lambda k|x| \leq CkM^{\alpha-1} = o(k \log M)$ and $\lambda k^2 \leq CkM^{-1/3} = o(k \log M)$. Hence the principal contribution is summable.

The truncation-remainder part of theorem 5.1 contributes

$$\sum_{k=0}^L \left(\frac{|x|}{R} \right)^k e^{-c(L-k)},$$

which is bounded by theorem 6.1.

In the boundary range $N < C_*(1 + |x|)$, the bound $|x| \leq \rho M^\alpha = o(M^{2/3})$ gives

$$k = L - N \geq AM^{2/3}/2$$

for all sufficiently large M , after adjusting the harmless constant in $L = \lfloor AM^{2/3} \rfloor$. Hence

$$\left(\frac{|x|}{R} \right)^k \leq \exp(-cM^{2/3} \log M).$$

The number of such boundary indices is

$$\#\{k : L - k < C_*(1 + |x|)\} = O(1 + |x|) = O(M^\alpha).$$

Thus the total boundary contribution is bounded by

$$O(M^\alpha) \exp(CM^\alpha) \exp(-cM^{2/3} \log M) = o(1),$$

because $\alpha < 2/3$. This proves the proposition. □

7 Hermite reconstruction and ratios

Define

$$\mathcal{H}_m(X) = H_m^{\text{phys}}(X/2),$$

so that

$$e^{Xt-t^2} = \sum_{m=0}^{\infty} \mathcal{H}_m(X) \frac{t^m}{m!}.$$

Write

$$E_M^{(L)}(t) = \sum_{m=0}^{\infty} \eta_m(M) \frac{t^m}{m!}.$$

Lemma 7.1 (Zeroth coefficient). *One has $\eta_0(M) = 1$.*

Proof. Since $E_M^{(L)}(0) = A_M^{(L)}(0) = S(0; M)$, and $S(0; M) = 1$, the claim follows. $\square$

Lemma 7.2 (Cauchy coefficient bound). *If $\sup_{|t| \leq r_M} |E_M^{(L)}(t)| \leq C_\theta$, then*

$$|\eta_m(M)| \leq C_\theta m! r_M^{-m}.$$

Proof. This is Cauchy's coefficient estimate. $\square$

Lemma 7.3 (Hermite reconstruction). *The polynomial in theorem 3.5 has the expansion*

$$\tilde{J}_\gamma^{d,n}(X) = \sum_{m=0}^d \binom{d}{m} \eta_m(M) \mathcal{H}_{d-m}(X).$$

Proof. Multiply the generating functions for e^{Xt-t^2} and $E_M^{(L)}(t)$ and take the coefficient of t^d . $\square$

Let $X_1 < \dots < X_{d-1}$ be the extrema of $\mathcal{H}_d$, and put $x_i = X_i/2$. Then x_i is a zero of H_{d-1}^{phys} .

Lemma 7.4 (Main-range Hermite ratio). *There are constants $C > 0$, $0 < \eta < 1$ such that, for $1 \leq m \leq \eta d$,*

$$\left| \frac{\mathcal{H}_{d-m}(X_i)}{\mathcal{H}_d(X_i)} \right| \leq \left(\frac{C}{\sqrt{d}} \right)^m \exp \left(C \frac{m^2}{d} \right).$$

Proof. Set

$$R_m(i) = \frac{H_{d-m}^{\text{phys}}(x_i)}{H_d^{\text{phys}}(x_i)}.$$

Then $R_0 = 1$, $R_1 = 0$, and the physical Hermite recurrence gives

$$R_{m+1} = \frac{2x_i R_m - R_{m-1}}{2(d-m)}.$$

Using $|x_i| \leq C\sqrt{d}$, and choosing $0 < \eta < 1/2$, we get for $m \leq \eta d$

$$|R_{m+1}| \leq \frac{C}{\sqrt{d}} |R_m| + \frac{C}{d} |R_{m-1}|.$$

We prove by induction that

$$|R_m| \leq \left(\frac{C_1}{\sqrt{d}} \right)^m \exp \left(C_1 \frac{m^2}{d} \right), \quad 0 \leq m \leq \eta d,$$

with C_1 sufficiently large. The cases $m = 0, 1$ are immediate. Assume the estimate holds for m and $m - 1$. Then

$$|R_{m+1}| \leq \frac{C}{\sqrt{d}} |R_m| + \frac{C}{d} |R_{m-1}|.$$

The first term is bounded by

$$\frac{C}{C_1} \left(\frac{C_1}{\sqrt{d}} \right)^{m+1} \exp \left(C_1 \frac{m^2}{d} \right),$$

and the second by

$$\frac{C}{C_1^2} \left(\frac{C_1}{\sqrt{d}} \right)^{m+1} \exp \left(C_1 \frac{(m-1)^2}{d} \right).$$

Choosing C_1 sufficiently large, these two contributions are absorbed by

$$\left(\frac{C_1}{\sqrt{d}} \right)^{m+1} \exp \left(C_1 \frac{(m+1)^2}{d} \right),$$

which closes the induction. Since $\mathcal{H}_{d-m}(X_i) = H_{d-m}^{\text{phys}}(x_i)$, the stated estimate follows. $\square$

7.1 A crude Hermite-function envelope

Let

$$\psi_n(x) = (2^n n! \sqrt{\pi})^{-1/2} e^{-x^2/2} H_n^{\text{phys}}(x)$$

be the normalized Hermite function.

Lemma 7.5 (Uniform upper envelope). *There is an absolute constant $C > 0$ such that*

$$\|\psi_r\|_\infty \leq C(r+1)^{1/4}.$$

Proof. The functions ψ_r satisfy

$$-\psi_r''(x) + x^2 \psi_r(x) = (2r+1) \psi_r(x), \quad \|\psi_r\|_2 = 1.$$

Thus $\|\psi_r'\|_2 \leq (2r+1)^{1/2}$. The one-dimensional Sobolev inequality

$$\|f\|_\infty^2 \leq 2\|f\|_2\|f'\|_2$$

gives the claim. $\square$

Lemma 7.6 (Standard Gauss–Hermite facts). *Let x_i be a zero of H_{d-1}^{phys} , and let $w_i^{(d-1)}$ be the corresponding Gauss–Hermite quadrature weight for the weight e^{-x^2} . Then*

$$w_i^{(d-1)} > 0, \quad \sum_i w_i^{(d-1)} = \sqrt{\pi},$$

$$|x_i| \leq \sqrt{2d-1},$$

and

$$w_i^{(d-1)} = \frac{2^{d-2}(d-1)!\sqrt{\pi}}{(d-1)^2 H_{d-2}^{\text{phys}}(x_i)^2}.$$

Proof. These are the standard Gauss–Hermite quadrature weight formula and the standard location bound for the zeros of physicists' Hermite polynomials; see, for instance, Szegő [6, Chapter VI] or DLMF [7, Chapter 18]. $\square$

Lemma 7.7 (Exponential lower bound at extrema). *If x_i is a zero of H_{d-1}^{phys} , then*

$$|\psi_d(x_i)| \geq e^{-Cd}.$$

Proof. By theorem 7.6,

$$w_i^{(d-1)} = \frac{2^{d-2}(d-1)!\sqrt{\pi}}{(d-1)^2 H_{d-2}^{\text{phys}}(x_i)^2}.$$

Since $H_{d-1}^{\text{phys}}(x_i) = 0$, the recurrence gives

$$H_d^{\text{phys}}(x_i) = -2(d-1)H_{d-2}^{\text{phys}}(x_i).$$

Therefore

$$H_d^{\text{phys}}(x_i)^2 = \frac{2^d(d-1)!\sqrt{\pi}}{w_i^{(d-1)}}.$$

Hence

$$\psi_d(x_i)^2 = (2^d d! \sqrt{\pi})^{-1} e^{-x_i^2} H_d^{\text{phys}}(x_i)^2 = \frac{e^{-x_i^2}}{d w_i^{(d-1)}}.$$

Using $w_i^{(d-1)} \leq \sqrt{\pi}$ and $|x_i|^2 \leq 2d$, we obtain

$$\psi_d(x_i)^2 \geq \frac{e^{-2d}}{d\sqrt{\pi}}.$$

The polynomial factor d^{-1} is absorbed into the exponential loss, so $|\psi_d(x_i)| \geq e^{-Cd}$. $\square$

Proposition 7.8 (Crude Hermite-function ratio at extrema). *For every zero x_i of H_{d-1}^{phys} and every $0 \leq r \leq d$,*

$$\left| \frac{\psi_r(x_i)}{\psi_d(x_i)} \right| \leq e^{Cd}.$$

Proof. Combine theorems 7.5 and 7.7. □

Lemma 7.9 (Tail Hermite ratio). *For $\eta d < m \leq d$,*

$$\left| \frac{\mathcal{H}_{d-m}(X_i)}{\mathcal{H}_d(X_i)} \right| \leq e^{Cd} d^{-m/2}.$$

This estimate is deliberately crude: the exponential loss e^{Cd} is harmless because the perturbative constant q_0 will be chosen exponentially small in the fixed parameter $1/\eta$ in theorem 8.1.

Proof. With $x_i = X_i/2$,

$$\frac{\mathcal{H}_{d-m}(X_i)}{\mathcal{H}_d(X_i)} = \frac{H_{d-m}^{\text{phys}}(x_i)}{H_d^{\text{phys}}(x_i)} = 2^{-m/2} \left(\frac{(d-m)!}{d!} \right)^{1/2} \frac{\psi_{d-m}(x_i)}{\psi_d(x_i)}.$$

By theorem 7.8, the last ratio is $\leq e^{Cd}$. For all $0 \leq m \leq d$, Stirling's lower bound gives

$$d^m \frac{(d-m)!}{d!} = \prod_{q=d-m+1}^d \frac{d}{q} \leq \frac{d^d}{d!} \leq e^d.$$

Thus

$$\left(\frac{(d-m)!}{d!} \right)^{1/2} \leq e^{d/2} d^{-m/2},$$

and the result follows. □

Proposition 7.10 (Hermite ratio package). *The estimates of theorems 7.4 and 7.9 hold uniformly for all extrema X_i and all $1 \leq m \leq d$.*

8 Hermite perturbation and Sturm comparison

Lemma 8.1 (Hermite perturbation). *For every fixed $\rho_t > 0$, there exists $c_\theta > 0$ such that, if*

$$\sqrt{d} \leq c_\theta M^\theta,$$

then

$$|\tilde{J}_\gamma^{d,n}(X_i) - \mathcal{H}_d(X_i)| < |\mathcal{H}_d(X_i)|$$

for every extremum X_i of $\mathcal{H}_d$, after increasing the lower bound on M for the finitely many small values of d .

Proof. By theorems 7.1 and 7.3,

$$\tilde{J}_\gamma^{d,n}(X) = \mathcal{H}_d(X) + \sum_{m=1}^d \binom{d}{m} \eta_m(M) \mathcal{H}_{d-m}(X).$$

By theorems 6.2 and 7.2, with $r_M = \rho_t M^\theta$,

$$|\eta_m(M)| \leq C_{\theta, \rho_t} m! r_M^{-m}.$$

Thus the relative perturbation at X_i is bounded by

$$C_{\theta, \rho_t} \sum_{m=1}^d \left(\frac{d}{\rho_t M^\theta} \right)^m \left| \frac{\mathcal{H}_{d-m}(X_i)}{\mathcal{H}_d(X_i)} \right|.$$

Set

$$q := \frac{C\sqrt{d}}{\rho_t M^\theta}.$$

First fix $\rho_t > 0$. Then C_{θ, ρ_t} is a finite constant. We choose $c_\theta > 0$ sufficiently small so that $\sqrt{d} \leq c_\theta M^\theta$ implies

$$q \leq q_0 := \frac{C c_\theta}{\rho_t},$$

where q_0 satisfies the two inequalities below.

For $m \leq \eta d$, theorem 7.4 gives

$$C_{\theta, \rho_t} q^m \exp(Cm^2/d) \leq C_{\theta, \rho_t} (q e^{C\eta})^m,$$

since $m^2/d \leq \eta m$. Choose q_0 so small that

$$C_{\theta, \rho_t} \sum_{m \geq 1} (q_0 e^{C\eta})^m < \frac{1}{4}.$$

This controls the main range.

For $m > \eta d$, theorem 7.9 gives terms bounded by

$$C_{\theta, \rho_t} e^{Cd} q^m.$$

Choose $q_0 < \exp(-2C/\eta)$, and then increase d_0 if necessary, so that

$$C_{\theta, \rho_t} e^{Cd} \frac{q_0^{\eta d}}{1 - q_0} < \frac{1}{4} \quad (d \geq d_0).$$

This controls the tail range for $d \geq d_0$.

It remains to cover the finitely many $d < d_0$. For each fixed d , the nonconstant coefficients in the normalized Hermite reconstruction tend to zero as $M \rightarrow \infty$ by the same Cauchy estimate, and hence the strict perturbation holds for all sufficiently large M . The finitely many resulting lower bounds for M are absorbed later by increasing C_ε . Therefore the total relative perturbation is strictly less than one. $\square$

Lemma 8.2 (Hermite extrema comparison). *Let P be a real polynomial of degree d with the same nonzero leading coefficient sign as $\mathcal{H}_d$. If*

$$|P(X_i) - \mathcal{H}_d(X_i)| < |\mathcal{H}_d(X_i)|$$

for all extrema X_i of $\mathcal{H}_d$, then P is hyperbolic.

Proof. At each extremum, P has the same sign as $\mathcal{H}_d$. These signs alternate, giving zeros in the intervals between consecutive extrema. Since the leading coefficient signs agree and are nonzero, P and $\mathcal{H}_d$ also have the same eventual signs at $\pm\infty$; the two exterior intervals supply the remaining zeros. Thus P has at least d real zeros, hence exactly d . $\square$

9 Affine transfer

Lemma 9.1 (Positivity of the xi-coefficients). *For every $m \geq 0$, one has $\gamma(m) > 0$.*

Proof. Griffin et al. [4, after eq. (1.1)] state that $\gamma(n) > 0$ for all $n \geq 0$, referring to the classical integral representation of the xi-function coefficients. See also O’Sullivan [5, Section 1]. We use only positivity and nonvanishing. $\square$

Lemma 9.2 (Real-affine transfer). *The Griffin normalization gives*

$$\tilde{J}_\gamma^{d,n}(X) = \frac{\gamma(M)^{d-1}}{\gamma(M-1)^d} \Delta(M)^d J_\gamma^{d,n} \left(\frac{\gamma(M-1)}{\gamma(M)} (\Delta(M)X - 1) \right).$$

Consequently, $J_\gamma^{d,n}$ is hyperbolic if and only if $\tilde{J}_\gamma^{d,n}$ is hyperbolic.

Proof. By theorem 9.1 and theorem 3.1, $\gamma(M), \gamma(M-1), \Delta(M) > 0$. Hence the displayed formula is a nonzero real scalar multiple of a nondegenerate real affine change of variable. Real-rootedness is preserved in both directions. $\square$

10 Proof of the main theorem

Proof of theorem 1.1. Fix $\varepsilon > 0$, and choose $\theta, \alpha, \delta, A, \rho_t, B$ as in theorem 2.1. Let c_θ be the constant supplied by theorem 8.1. Increase C_ε so that the range conditions in theorem 2.2 hold and so that the radius-to-scale implication

$$M \geq C_\varepsilon d^{3+\varepsilon} \implies \sqrt{d} \leq c_\theta M^\theta$$

holds.

By theorem 3.5, the analytic reconstruction is exactly the Griffin-normalized Jensen polynomial. By theorem 3.3, Rice’s estimate applies to F_M . By theorem 4.2, $A_M^{(L)}$ decomposes into shifted kernels. By theorem 5.1, these kernels satisfy the main and boundary estimates, including the truncation-remainder term. The bounded Hermite–Gaussian summation in theorem 6.2 gives

$$\sup_{|t| \leq \rho_t M^\theta} |e^{t^2} A_M^{(L)}(t)| \leq C_{\theta, \rho_t}.$$

Cauchy’s estimate gives

$$|\eta_m(M)| \leq C_{\theta, \rho_t} m! (\rho_t M^\theta)^{-m}.$$

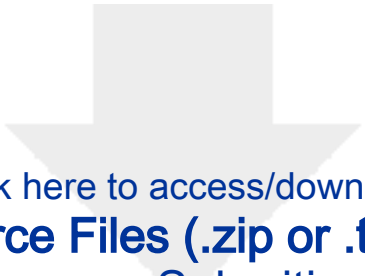
The choice of C_ε ensures $\sqrt{d} \leq c_\theta M^\theta$, so theorem 8.1 applies. Since $\eta_0(M) = 1$, the leading coefficient of $\tilde{J}_\gamma^{d,n}$ is exactly the leading coefficient of $\mathcal{H}_d$. Therefore theorem 8.2 shows that $\tilde{J}_\gamma^{d,n}$ is hyperbolic. Finally, theorem 9.2 transfers hyperbolicity to $J_\gamma^{d,n}$. It remains only to cover the finitely many small values of d excluded in the Hermite perturbation lemmas. For each such fixed exceptional degree d , the Cauchy estimate above gives $|\eta_m(M)| \rightarrow 0$ as $M \rightarrow \infty$ for every $1 \leq m \leq d$, so there is a threshold M_d such that the same strict extrema comparison holds whenever $M \geq M_d$. Enlarging C_ε , if necessary, so that $C_\varepsilon d^{3+\varepsilon} \geq M_d$ for all these finitely many exceptional degrees, the stated hypothesis forces $M \geq M_d$ in every remaining small-degree case. The finitely many lower bounds for M required by the asymptotic estimates are absorbed into the same enlargement of C_ε . $\square$

11 Endpoint problem

The proof requires $\theta < 1/6$. Equivalently, with $\alpha = 1/2 + \theta$, it requires the strict inequality $\alpha < 2/3$. This strict subcritical gap is used in the shifted-saddle estimates, for example in absorbing terms such as $M^{3\alpha-2}$, and in the boundary summation where $M^{2/3} \log M$ must dominate M^α . The endpoint $\theta = 1/6$, corresponding to $M \asymp d^3$, is not covered by the present argument.

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